Climate Change Impacts to Hydrology

ENVS 423L/EPS 522, Fall 2025

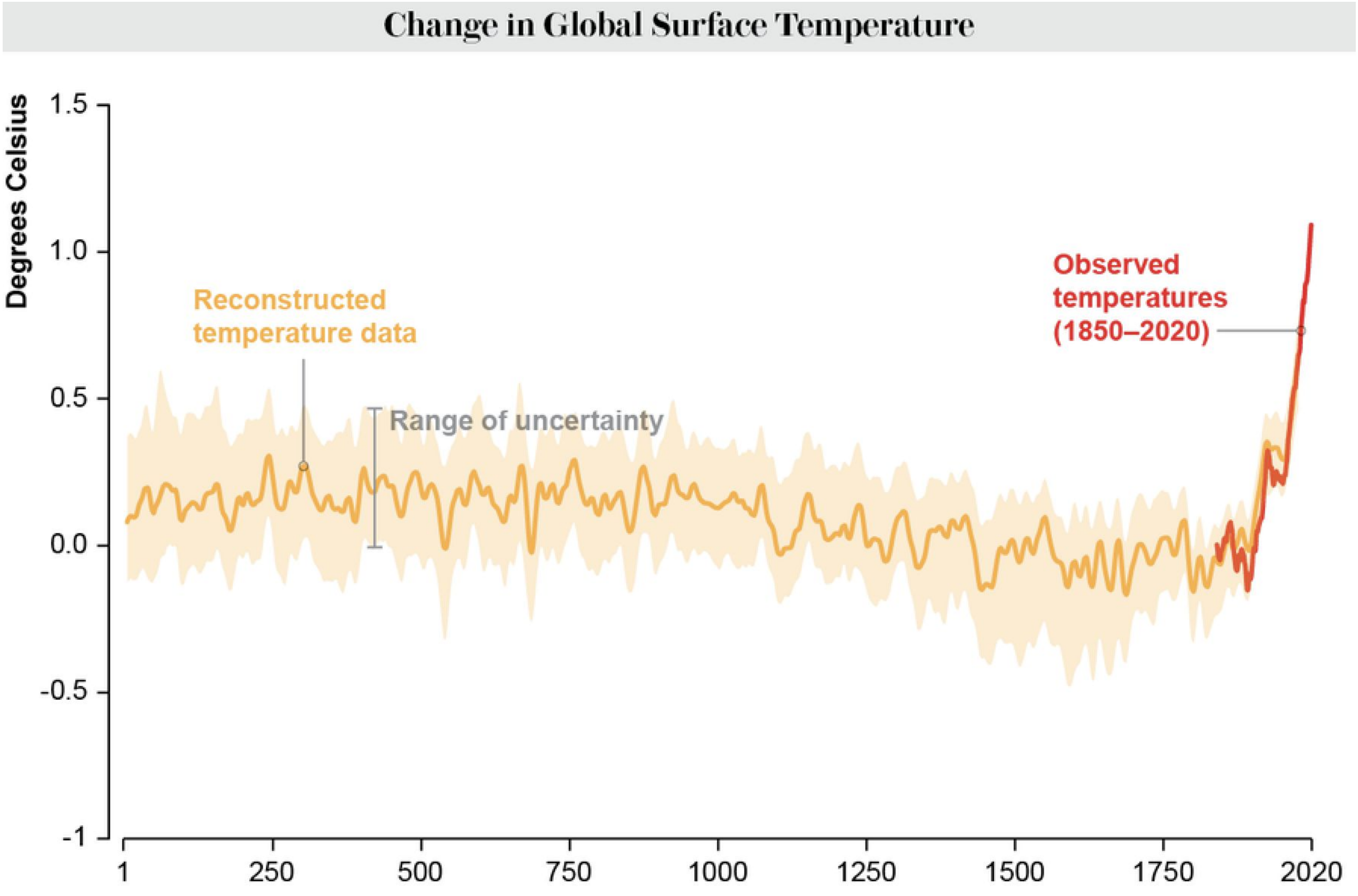
Thursday, November 13

Housekeeping

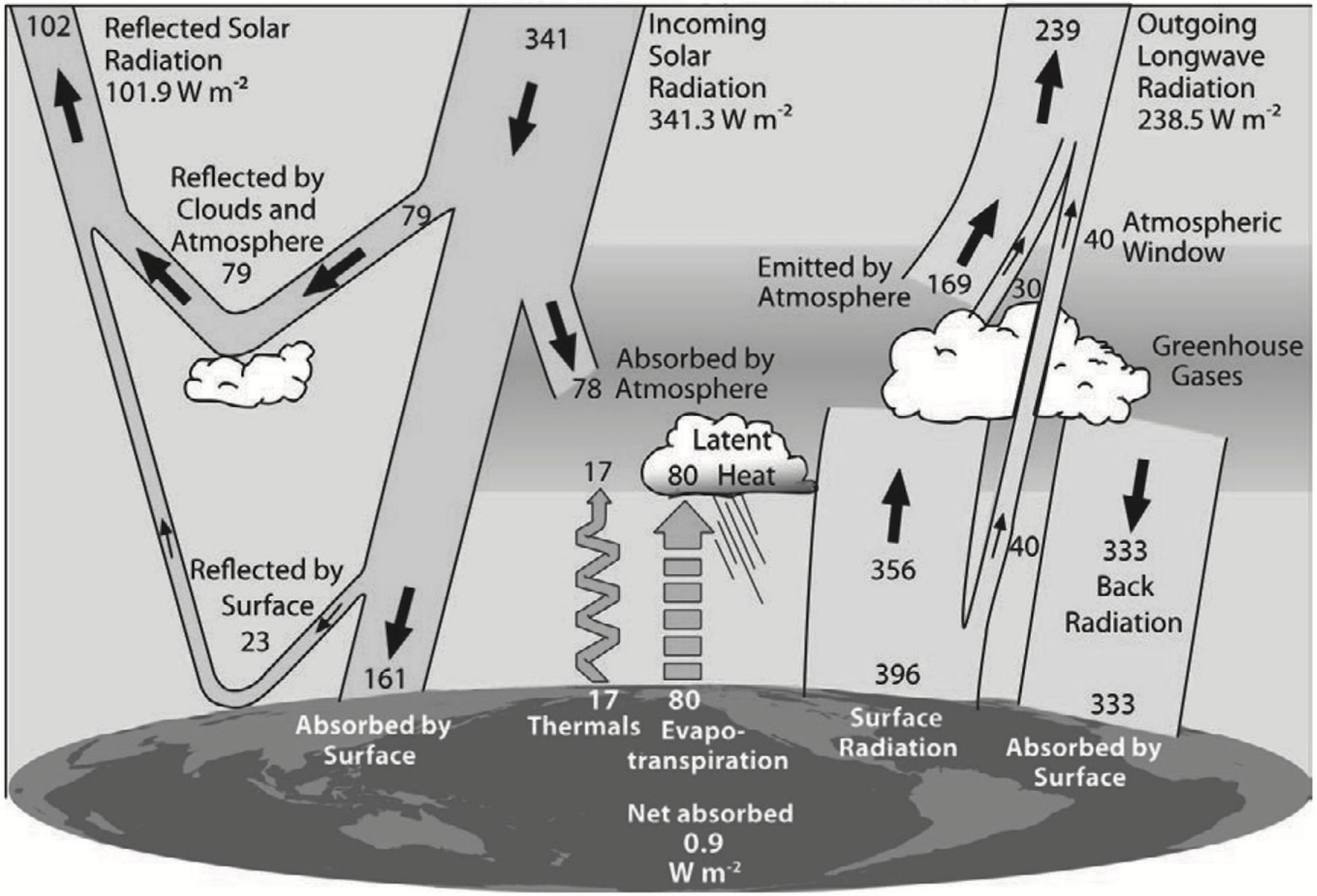
- Job ad for MSc/PhD position in my lab starting Fall 2026
- Homework 7 deadline pushed to November 18
- Tech memo rubric posted on GitHub
- Homework 8 will be an in-class quiz during one of the classes next week

IPCC Working Group 6 Synthesis Report

“Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming, with global surface temperature reaching 1.1°C above 1850-1900 in 2011-2020.”

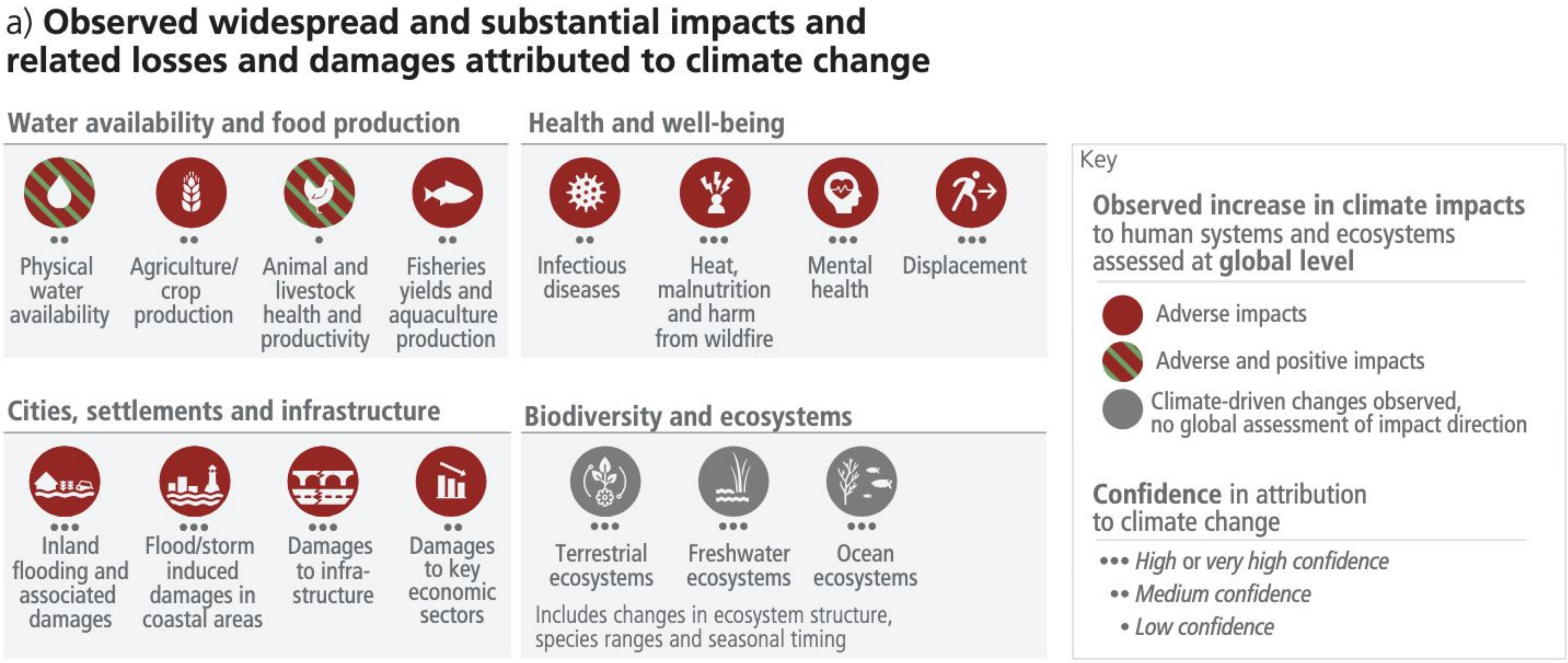


Greenhouse Effect



IPCC Working Group 6 Synthesis Report

“Widespread and rapid changes in the atmosphere, ocean, cryosphere and biosphere have occurred.”



IPCC Working Group 6 Synthesis Report

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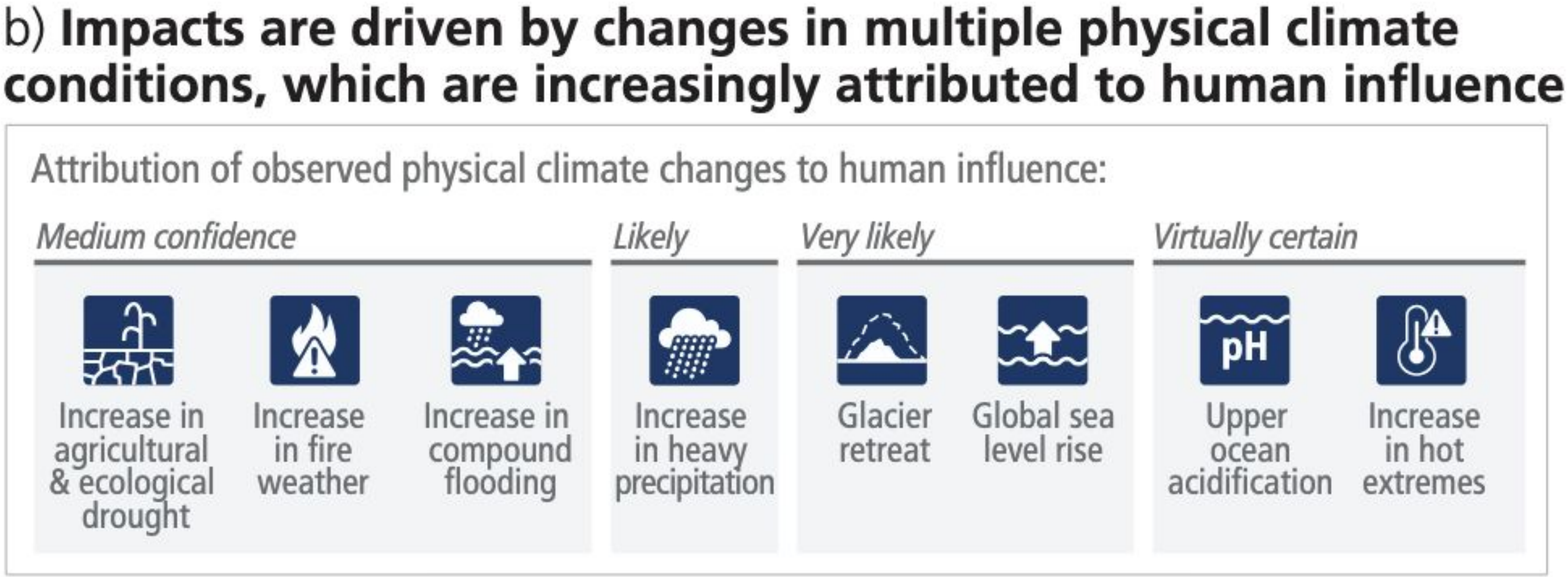
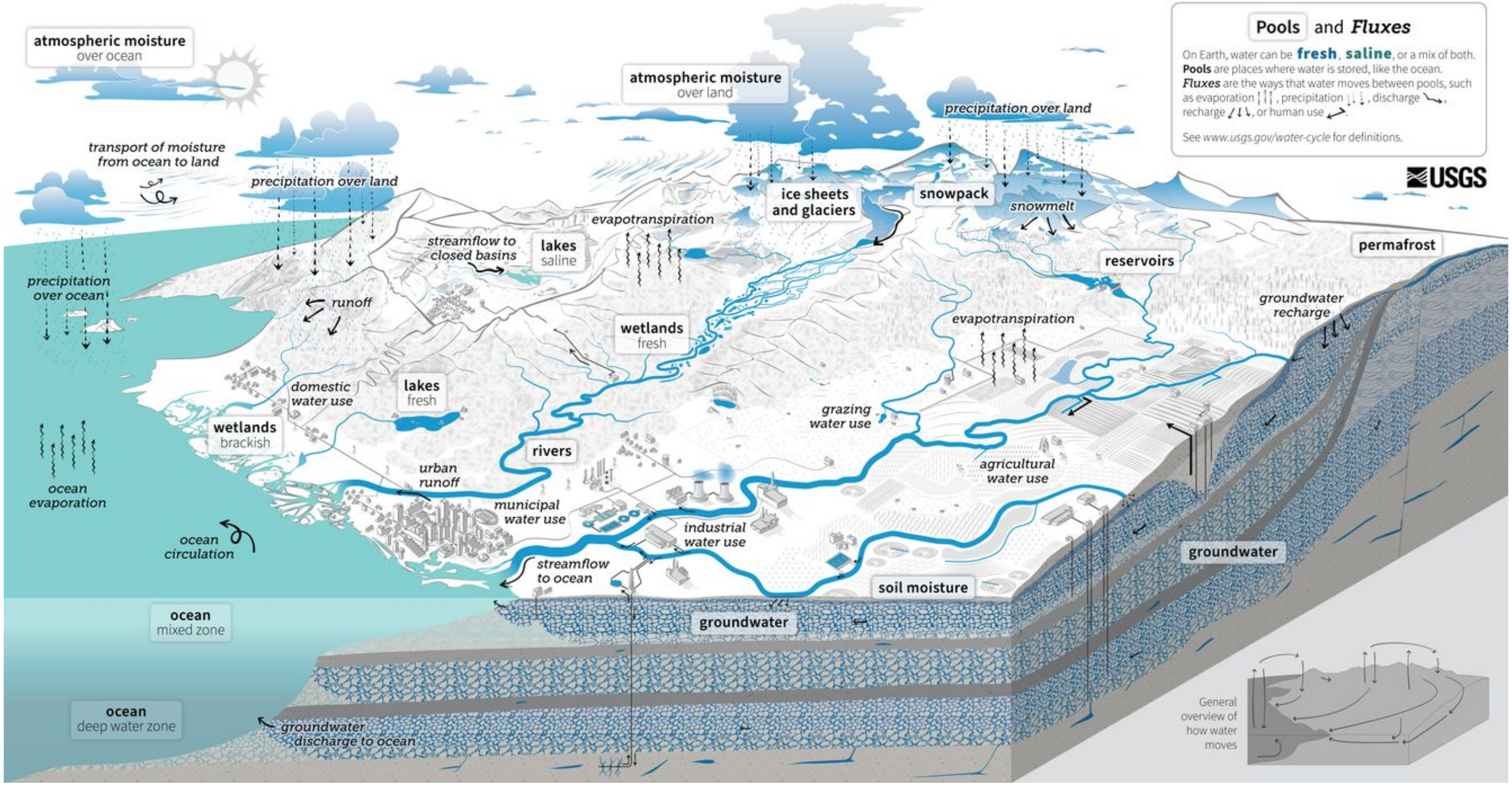


Intensification of Hydrologic Cycle



Atmospheric water vapor increases 7% per ℃ (Clausius-Clapeyron equation, warm air holds more water).
Warmer ocean surface temperature leads to more evaporation, and more water vapor leads to more precipitation.

Detecting, attributing, and projecting impacts to regional hydrology are challenging for several reasons:

- Precipitation, the main driver of terrestrial hydrology, is affected natural climate variability (ENSO, PDO, AMO, etc.) as well as anthropogenic climate change
- General circulation (ocean-atmosphere) models are coarse, do not resolve cloud processes (important for convective precipitation), and do not simulate terrestrial hydrology
- ET, largest component of water balance after precipitation, exhibits complex feedbacks with CO2 concentrations, warmer temperatures, growing season length, and vegetation shifts/disturbances.
- Even in the absence of climate change, humans can substantially alter hydrologic systems, and including human-climate-hydrology feedbacks is very challenging in predictive models

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In-class Activity

- Form 5 groups
- Claim one of the following topics:
  1. Precipitation (including snow)
  2. Evapotranspiration
  3. Streamflow
  4. Vadose zone
  5. Groundwater
- Based on what you know about this component of the hydrologic cycle, develop some hypotheses of how you think it is shifting in response to climate change

In-class Activity

- Conduct a rapid (~30 minute) literature review to determine the current scientific consensus of how climate change is impacting/will impact this store/flux
- Summarize findings/figures in Google slides, include a slide at the end with references
- Finish slides by 10:15, and each group will have 5 min to present their findings, with 5 min at the end for a discussion

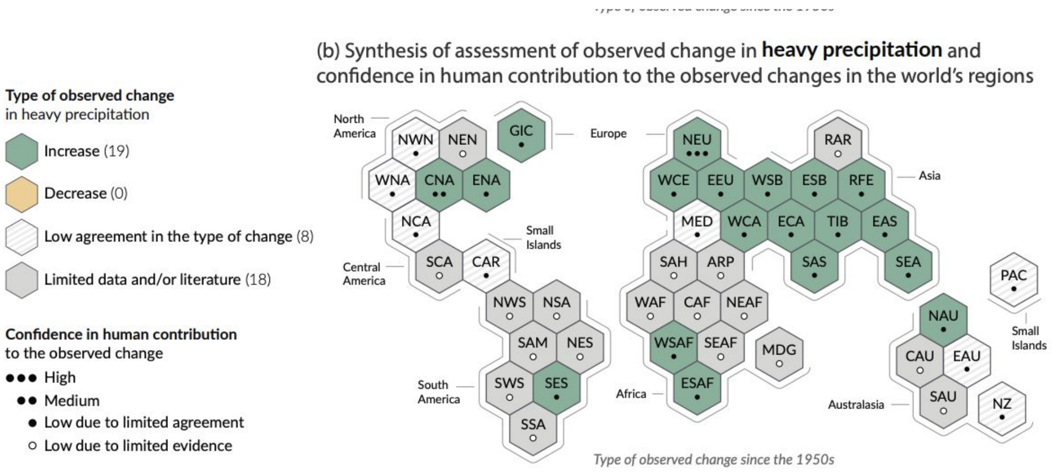
Considerations

- Have impacts already been detected/attributed?
- What are the forecasted impacts?
  - Near-term versus long-term? (i.e., increased runoff due to alpine glacier melt until glaciers are gone…)
- Threshold/tipping point behavior based on emissions pathway
  - i.e., AMOC shutdown
- Spatial differences in impacts
  - i.e., wet gets wetter, dry gets drier**
- Interactions with other fluxes/stores
  - i.e., streamflow variability leading to greater reliance on groundwater extraction

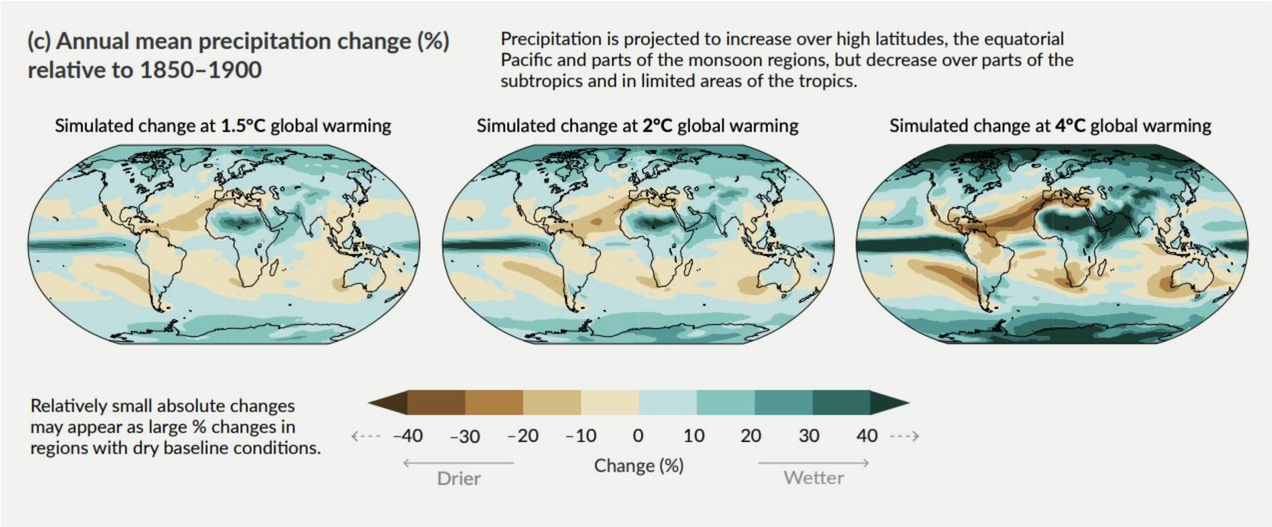
Precipitation

By: Andrew G, Heather FG, Matt P, Ezekiel M

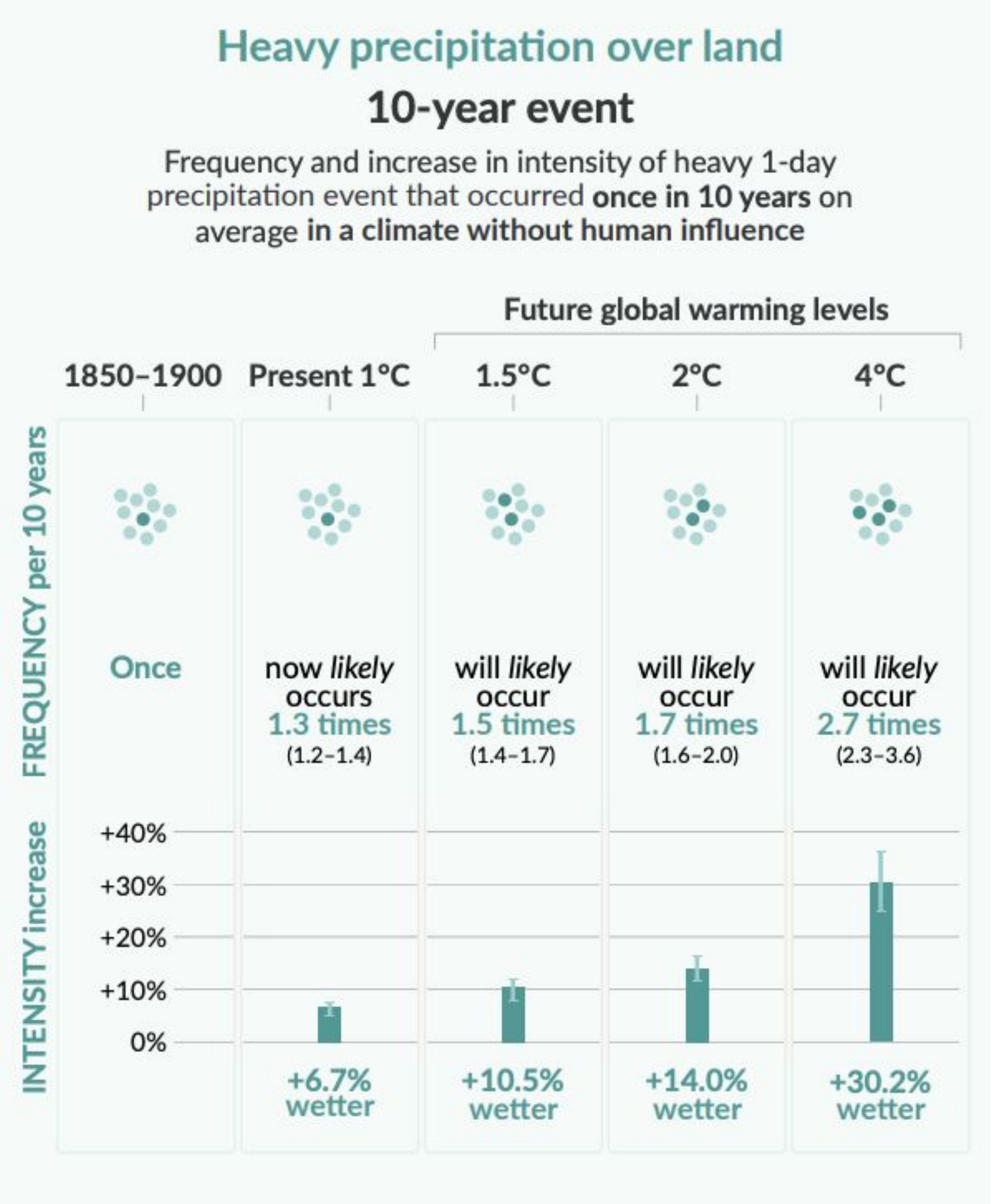
Hypothesis:



(b) Synthesis of assessment of observed change in **heavy precipitation** and confidence in human contribution to the observed changes in the world's regions
Type of observed change in heavy precipitation
Increase (19)
Decrease (0)
Low agreement in the type of change (8)
Limited data and/or literature (18)
Confidence in human contribution to the observed change
●●● High
●● Medium
● Low due to limited agreement
○ Low due to limited evidence
North America
Europe
Asia
Central America
Small Islands
South America
Africa
Australasia
NWN NEN GIC WNA CNA ENA NCA SCA CAR NWS NSA SAM NES SWS SES SSA
NEU RAR WCE EEU WSB ESB RFE MED WCA ECA TIB EAS SAH ARP SAS SEA WAF CAF NEAF WSAF SEAF MDG ESAF
PAC NAU CAU EAU SAU NZ
Type of observed change since the 1950s



(c) Annual mean precipitation change (%) relative to 1850–1900
Precipitation is projected to increase over high latitudes, the equatorial Pacific and parts of the monsoon regions, but decrease over parts of the subtropics and in limited areas of the tropics.
Simulated change at **1.5°C** global warming
Simulated change at **2°C** global warming
Simulated change at **4°C** global warming
Relatively small absolute changes may appear as large % changes in regions with dry baseline conditions.
−40 −30 −20 −10 0 10 20 30 40
Change (%)
Drier
Wetter



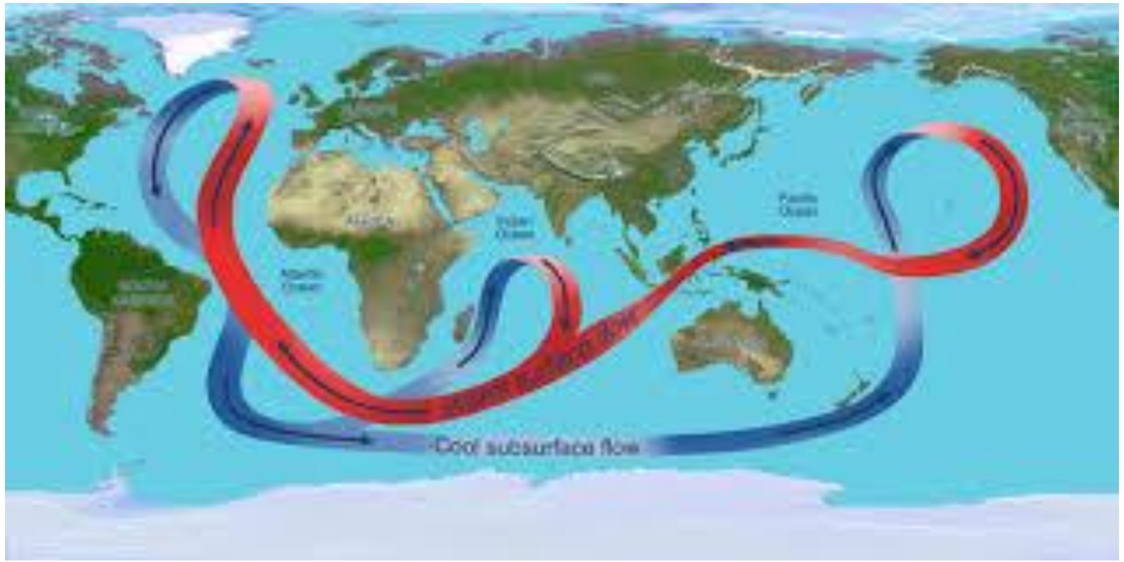
Heavy precipitation over land
10-year event
Frequency and increase in intensity of heavy 1-day precipitation event that occurred **once in 10 years** on average **in a climate without human influence**
Future global warming levels
1850–1900
Present 1°C
1.5°C
2°C
4°C
FREQUENCY per 10 years
Once
now *likely* occurs 1.3 times (1.2–1.4)
will *likely* occur 1.5 times (1.4–1.7)
will *likely* occur 1.7 times (1.6–2.0)
will *likely* occur 2.7 times (2.3–3.6)
INTENSITY increase
+40%
+30%
+20%
+10%
0%
+6.7% wetter
+10.5% wetter
+14.0% wetter
+30.2% wetter

Detected/Attributed Effects

- Decreased snowpack
- Increased intensity/frequency of precipitation events
- Water quality issues (Bad things in water)
- Drought/Wildfires

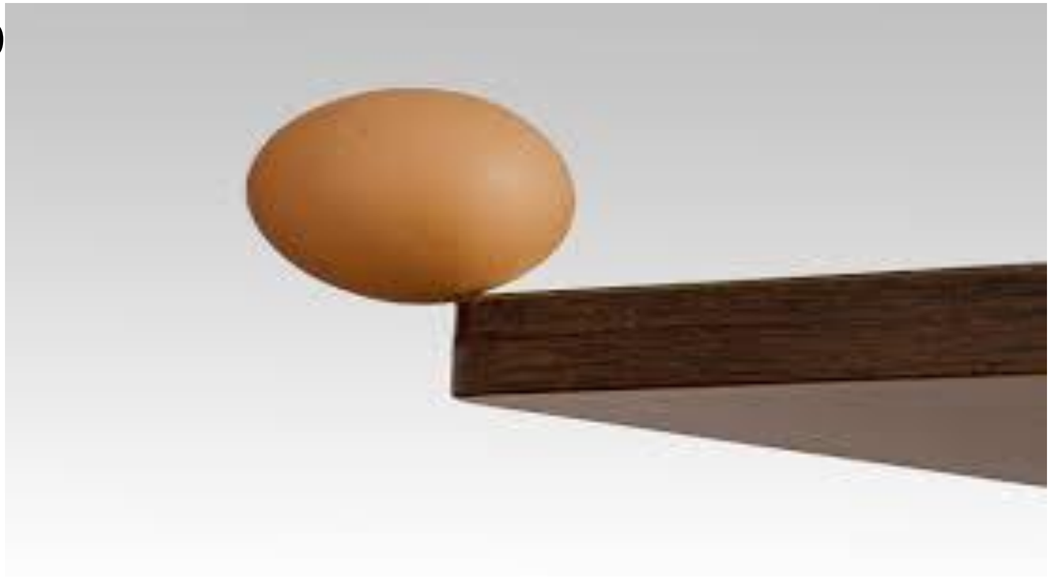
Forecasted Impacts

- Disruption of pacific atmospheric rivers
- Precipitation disruption for west coast
- Changes to tropical monsoons
- Great impact on Tropical rainforests such as Amazon
- Regional wetting/drying



Threshold/Tipping Point

- 1.5 Degrees C
- More frequent droughts and shifts in regional water supplies
- Losing rainforests(Amazon) causes reduces ET, causing positive feedback loop and leading to climatic tipping point
- 56.8 billion metric tons CO2
- Summer monso



Spatial Differences

Storm affected areas are likely to experience increases in precipitation and increased risk of flooding.

Dry areas become more dry.

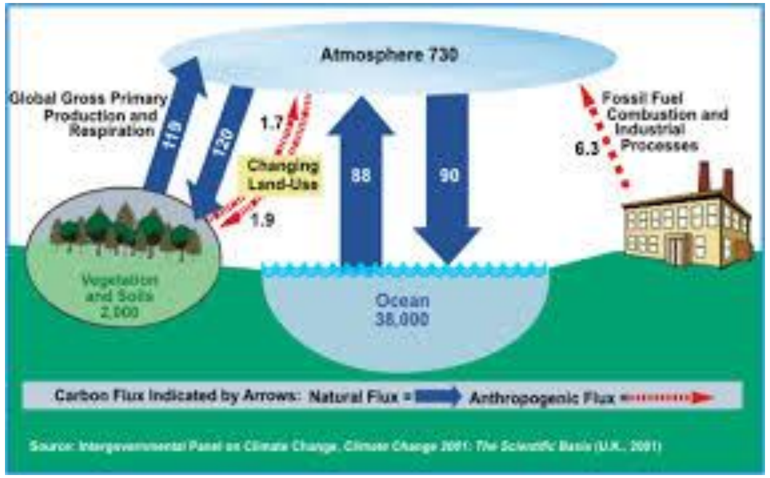
Interactions with other Fluxes

Increased evaporation allows for more water to be in the atmosphere.

Increased precip in some areas means more ground water.

In other areas where less precip occurs more droughts and less groundwater.

Increased flooding due to increased rain increases streamflow.



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IPCC. *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Core Writing Team, H. Lee and J. Romero, eds., IPCC, Geneva, Switzerland, 2023, pp. 35-115, doi:10.59327/IPCC/AR6-9789291691647.

Lenton, Timothy M., et al., editors. *The Global Tipping Points Report 2025*. University of Exeter, 2025.

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ET Changes with Global Warming

Bidek Pokharel, Elliott Robinson, Adam Splitek

Hypothesis Draft 0

- Evap will increase overall with increasing temp
    - The other climatic factors from global warming will also increase evap.
- Transpiration will increase from global warming due to increased perci
percip being liquid,
- As a result, Groundwater loss will increase in part due to the change i
Groundwater recharge may not change or may slow due to increased
plants.

Hypothesis

- As global warming increases, we will see an increase in water use effi
content, vapor pressure deficits, plant physiological changes, increase
some areas, and decreases in regional ET in some areas. Thus, we w
changes in global ET, of around ~1% per degree C° warming, as oppo
major increases or decreases in ET.

https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021EF002221

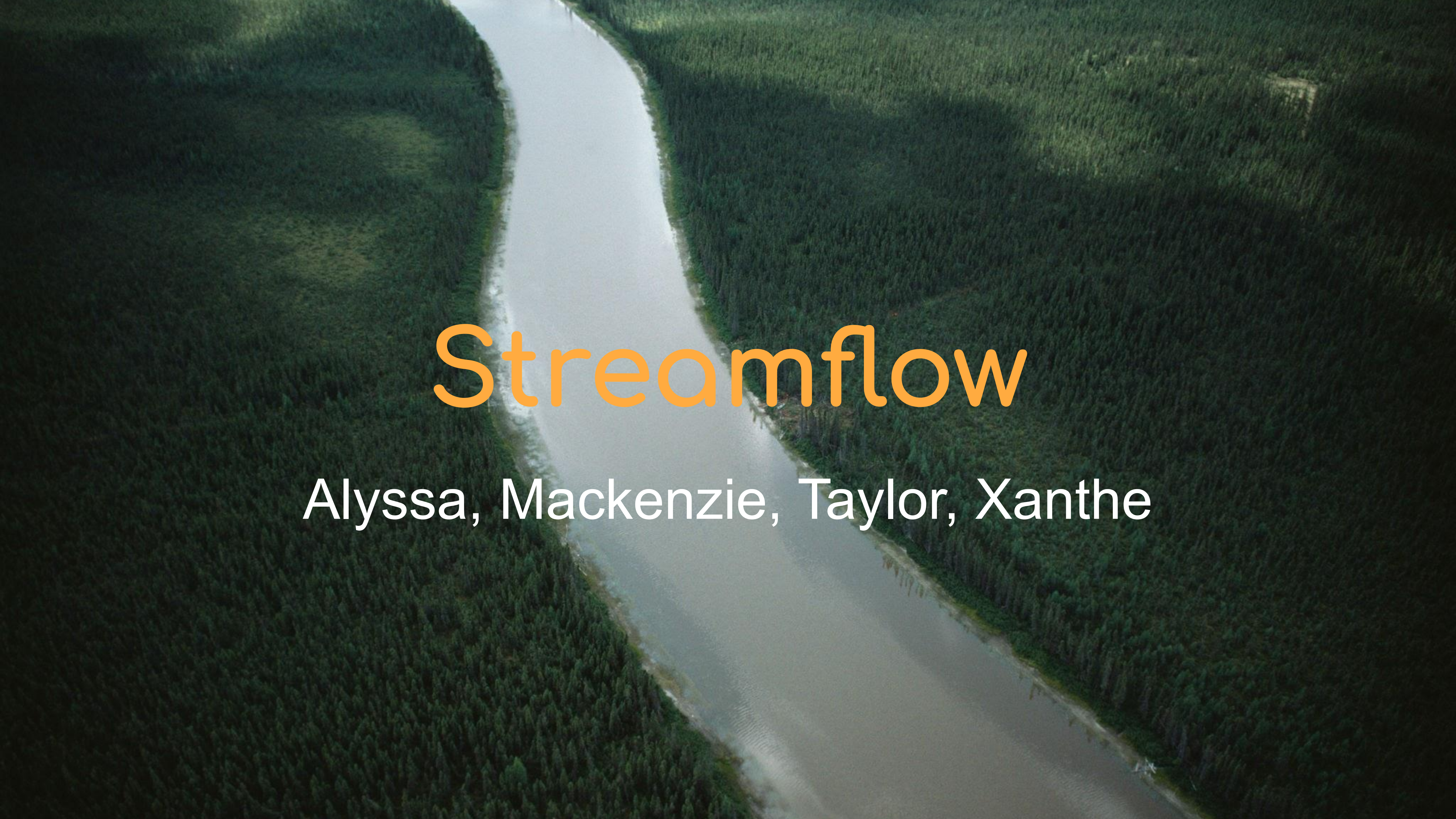
Elliott Robinson and Adam Splitek's findings

- Warming increases evaporative demand
  - Areas that have enough water will see an increase in evap
  - Areas that don't have enough water will not see a significant increase in actual evap (read southwest)
  - https://www.nature.com/articles/s41467-020-14688-0
- Many changes in ET balance out.
  - Land use changes is the best example. While some areas see an increase or decrease in ET due to land use changes, global ET balances out form this factor.
  - https://www.sciencedirect.com/science/article/pii/S016819232100349X#sec0016
  - Models of ET done at the regional scale show increases in some simulations/areas and decreases in others, reflecting a lot of uncertainty: https://doi.org/10.1029/2010GL046230
- Plant physiological changes moderate the degree to which changes in potential ET are realized as actual ET
  - https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2021EF002221
- Possible increase in evaporation from human biological processes (perspiration, respiration): https://doi.org/10.3390/ijerph19148638

Bidek Pokharel's findings

- WUE and SWC will increase as vapor pressure deficit increases
- Transpiration decreases with increased in VPD in ecosystems.
- WUE and SWC is largely effected due to the transpiration/evapotranspiration rate.

https://www.sciencedirect.com/science/article/pii/S0168192321002914



Streamflow

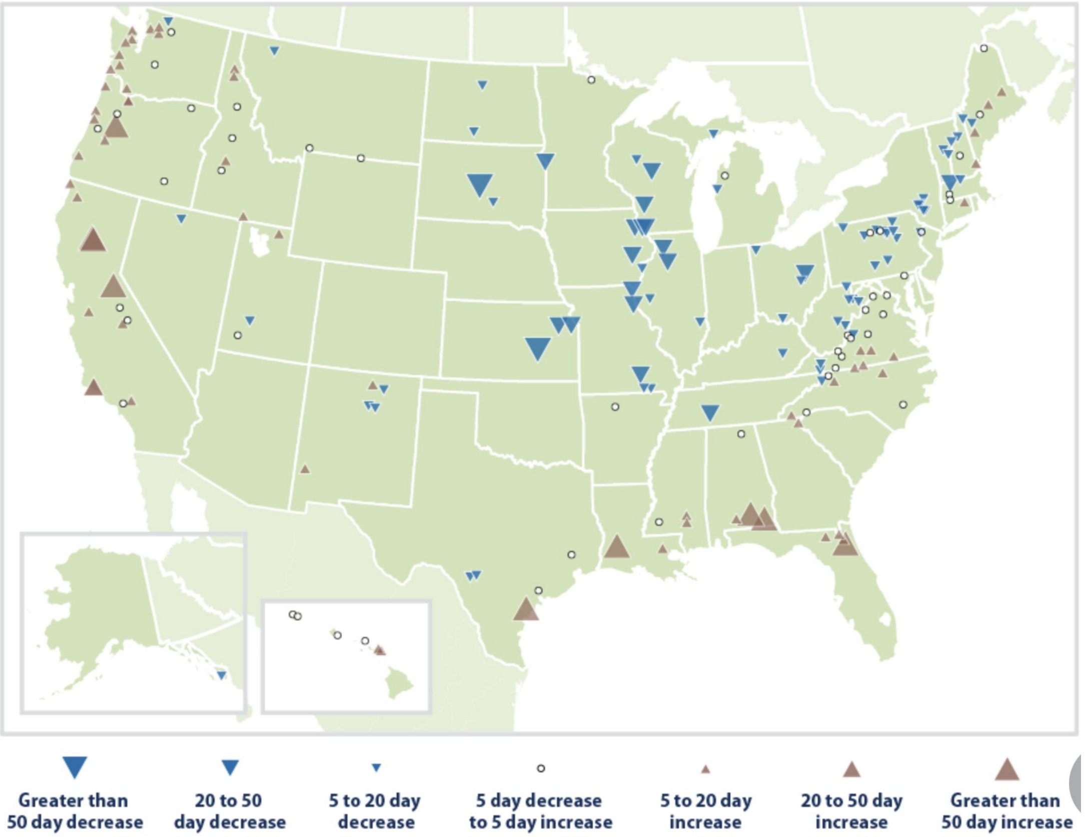
Alyssa, Mackenzie, Taylor, Xanthe

Hypothesis

As climate change continues to be a prevalent issue, streamflow will become more variable across the globe because off reduction in snowpack, more intense precipitation storms, flooding, prolong drought, and timing of flow will change.

Have impacts already been detected/attributed? What are the forecasted impacts? Near and long term?

- Winter-spring runoff has shifted by occurring 2 to 10 days earlier than what has been historically recorded.
- In locations that are typically more arid there has been an increase in low streamflow days by by 10-50 days and in typically more wet locations there has been a decrease of high streamflow days. by 5-50 days.
- Forecasted Impacts:
  - Lower streamflow - Near Term
  - Earlier and shorten runoff seasons - Long Term



Number of Days with Very Low Streamflow, 1940–2022
Source:https://www.epa.gov/climate-indicators/climate-change-indicators-streamflow#tab-3

Threshold/tipping point behavior based on emissions pathway

- Loss of glaciers and mountain snowpack
  - *Islam, F. A. S. (2025). Global Impact of Climate Change: Glacial Melt, Sea Level Rise, Water Salinization and Emergent Pathogen Risks. Asian Journal of Environment & Ecology, 24(5), 91–113. https://doi.org/10.9734/ajee/2025/v24i5697*

- Precipitation effects
  - Changes in sea ice cover affecting AMOC
  - ENSO
  - *Wunderling, N., von der Heydt, A. S., Aksenov, Y., Barker, S., Bastiaansen, R., Brovkin, V., Brunetti, M., Couplet, V., Kleinen, T., Lear, C. H., Lohmann, J., Roman-Cuesta, R. M., Sinet, S., Swingedouw, D., Winkelmann, R., Anand, P., Barichivich, J., Bathiany, S., Baudena, M., … Willeit, M. (2024). Climate tipping point interactions and cascades: A review. Earth System Dynamics, 15(1), 41–74. https://doi.org/10.5194/esd-15-41-2024*

- Groundwater depletions from pumping & drought
  - *Barlow, P.M., and Leake, S.A., 2012, Streamflow depletion by wells—Understanding and managing the effects of groundwater pumping on streamflow: U.S. Geological Survey Circular 1376, 84 p. (Also available at http://pubs.usgs.gov/circ/1376/.)*

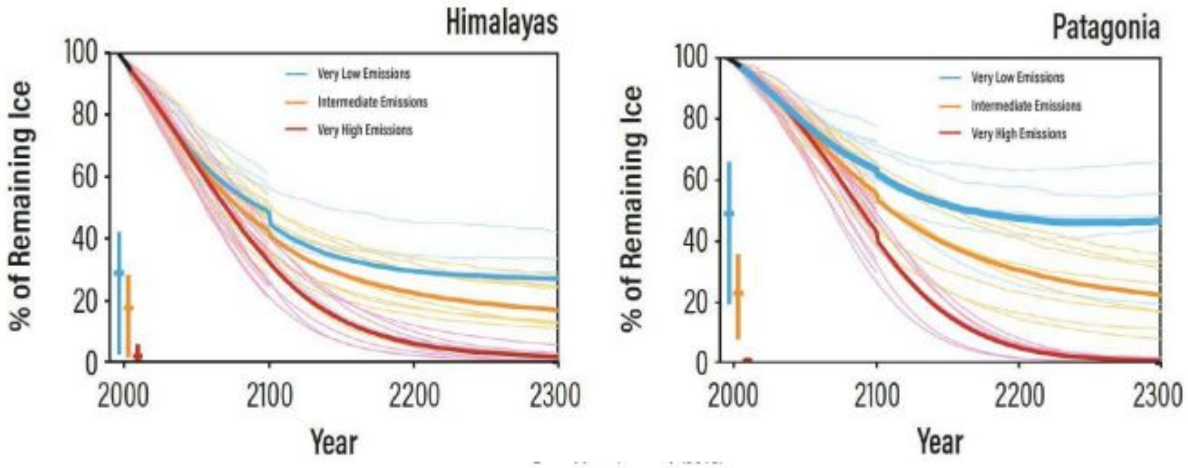


Fig. 1. Himalayan and Patagonian projection year versus percentage of remaining ice (Figure Source: (Marzeion et al., 2012, Marzeion et al., 2021)

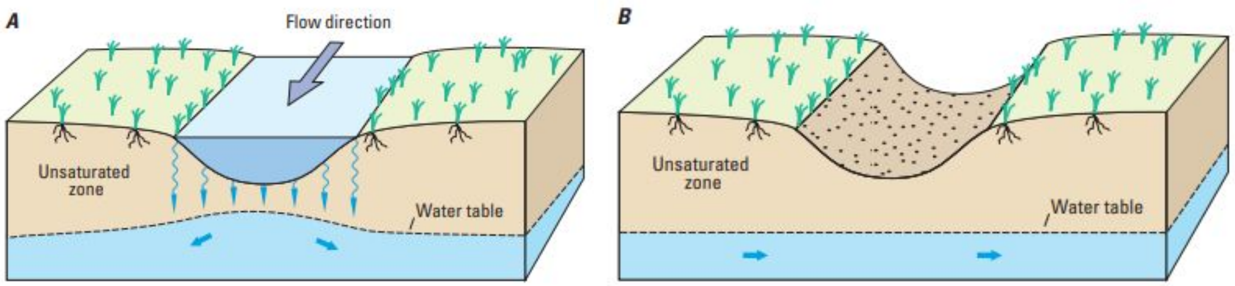


Figure 5. Disconnected stream reaches are separated from the groundwater system by an unsaturated zone. In *A*, streamflow is a source of recharge to the underlying groundwater system, but in *B*, streamflow and groundwater recharge have ceased (modified from Winter and others, 1998).

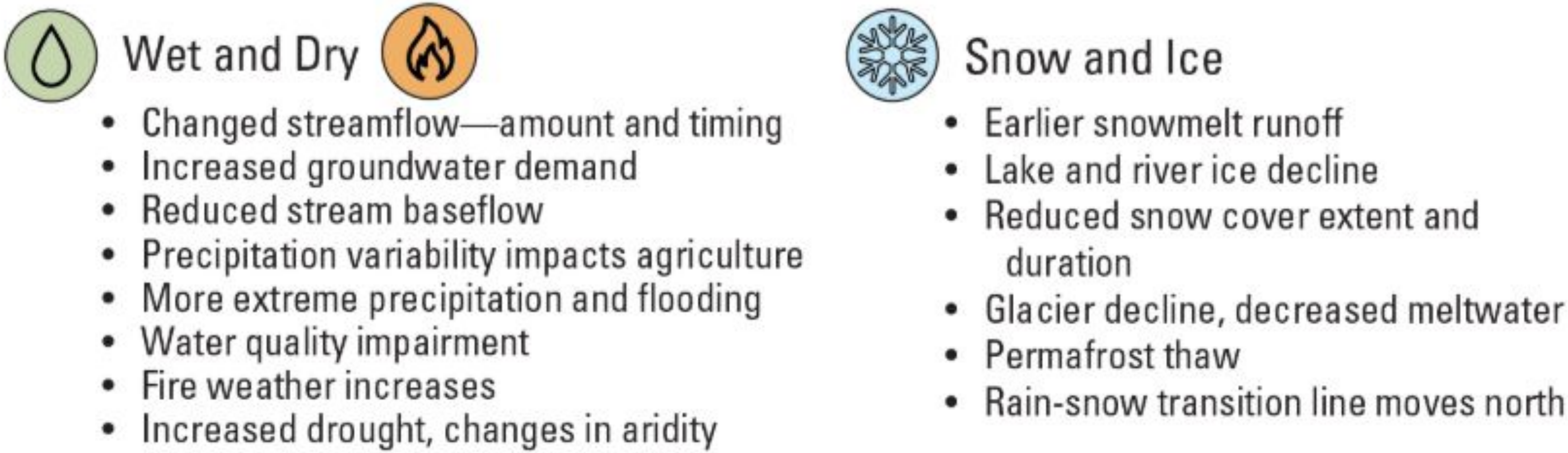
Spatial differences in impacts

•i.e., wet gets wetter, dry gets drier**

- Events like fires and floods will occupy larger physical areas which will impact streamflow
- Impacts like increase aridity and habit change will be observed at different latitudes and elevations (Mercier 2021)
- Crop deficiencies in already arid regions like Spain (Iglesias 2000)
- More than two-thirds of climate models project a significant increase in average annual runoff across almost a quarter of the land surface, and a significant decrease over 14% (Gosling 2021)

Interactions with other fluxes/stores

- Impacts will likely be highly variable
- Even within the same region, watershed impacts will vary



Wet and Dry

- Changed streamflow—amount and timing
- Increased groundwater demand
- Reduced stream baseflow
- Precipitation variability impacts agriculture
- More extreme precipitation and flooding
- Water quality impairment
- Fire weather increases
- Increased drought, changes in aridity

Snow and Ice

- Earlier snowmelt runoff
- Lake and river ice decline
- Reduced snow cover extent and duration
- Glacier decline, decreased meltwater
- Permafrost thaw
- Rain-snow transition line moves north

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Desert Research Institute. (2023, April 25). *Climate change is already impacting stream flows across the U.S.*https://www.dri.edu/streamflow_climate_change/

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Thee Vadose Zone

Rhett Sanders-Spencer, Elva Granados, Christopher O'Connor, William Mejia

Hypotheses

With climate change in arid snowmelt-driven regions, an increase in regional temperature would lead to:

1. higher evaporative demand (E)
2. shift in phase of precip from snow to rain

which would lead to less soil moisture in the vadose zone.

An increase in more intense rainfall events may lead to

1. erosion of the vadose zone
2. more crust formation on drier soils

which would lead to less soil moisture in the vadose zone.

Have impacts been detected / attributed?

- Deep drainage is less affected by snowmelt fraction since it happens over “longer time scales” (Hammond et al. 2019)
- “The conclusion is reached that significant changes in the composition of vadose zone pore waters should occur in response to climate-forced changes in infiltration flux and surface temperature” (Glassley et al., 2002)

Spatial differences in impacts

Vadose zone thickness is spatially heterogeneous, which affects groundwater recharge lag times.

References

H. Pfletschinger, K. Prömmel, C. Schüth, M. Herbst, I. Engelhardt; Sensitivity of Vadose Zone Water Fluxes to Climate Shifts in Arid Settings. *Vadose Zone Journal* 2014;; 13 (1): vzj2013.02.0043. doi: https://doi.org/10.2136/vzj2013.02.0043

https://www.sciencedirect.com/science/article/pii/S0022169414005915

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How Climate Change Affects Groundwater

Paloma Lopez, Katherine Jones, Ellie Lindsey, Marisa Davis

Thesis Statement:

Climate change increases changes in groundwater, resulting in dry places (like the southwest) continuing to get drier and more water strained while wet places (like the PNW) may experience some water strain or might get more groundwater recharge.

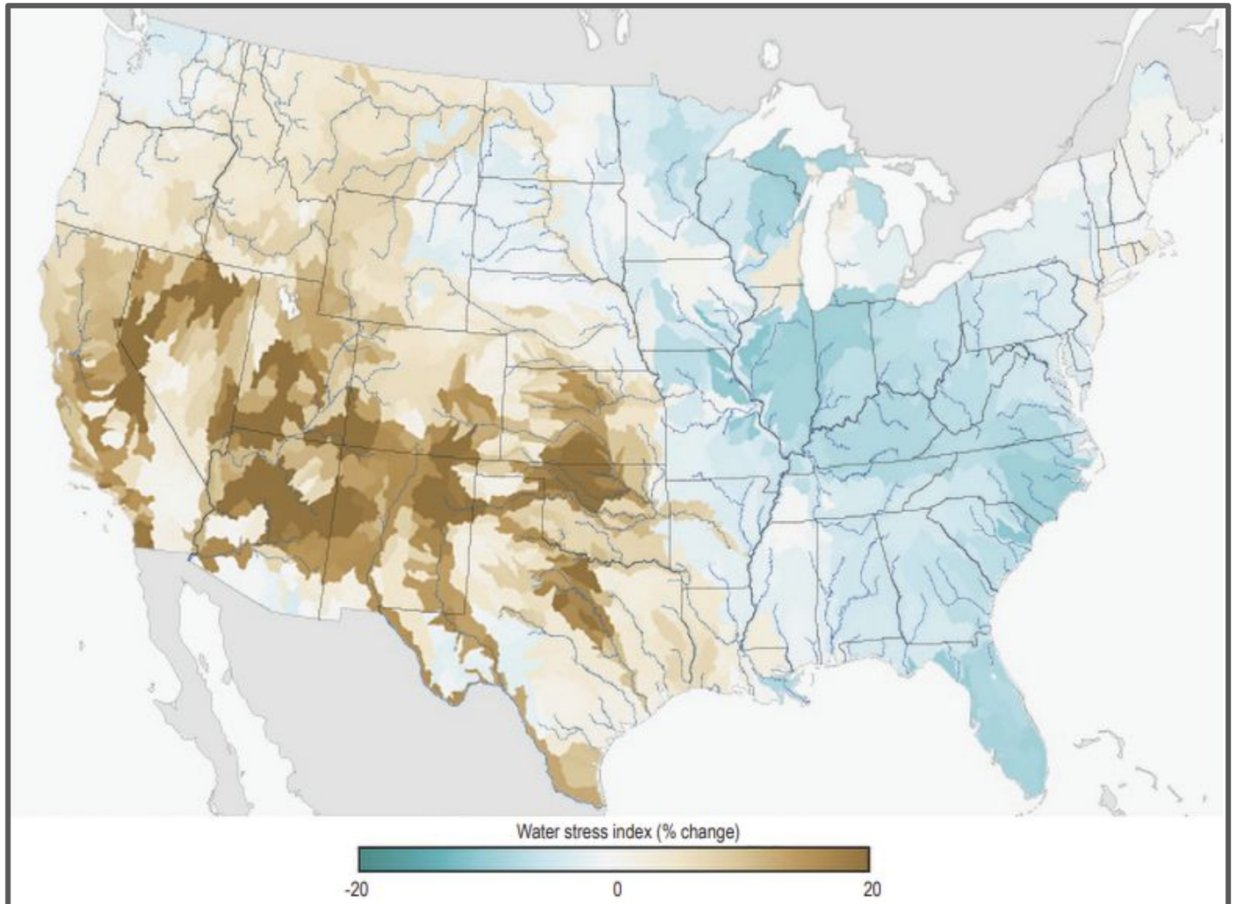
Global Overview

- warmer temperatures lead to
  - reduced aquifer recharge
  - decreased snowpack
- increased reliance on groundwater as surface water diminishes
- Increased pumping due to water scarcity
- Increased evapotranspiration: higher temps lead to more evaporation from soil and plants so more water is lost in the atmosphere before it can seep into the ground
- Reduced groundwater quality (warmer water)

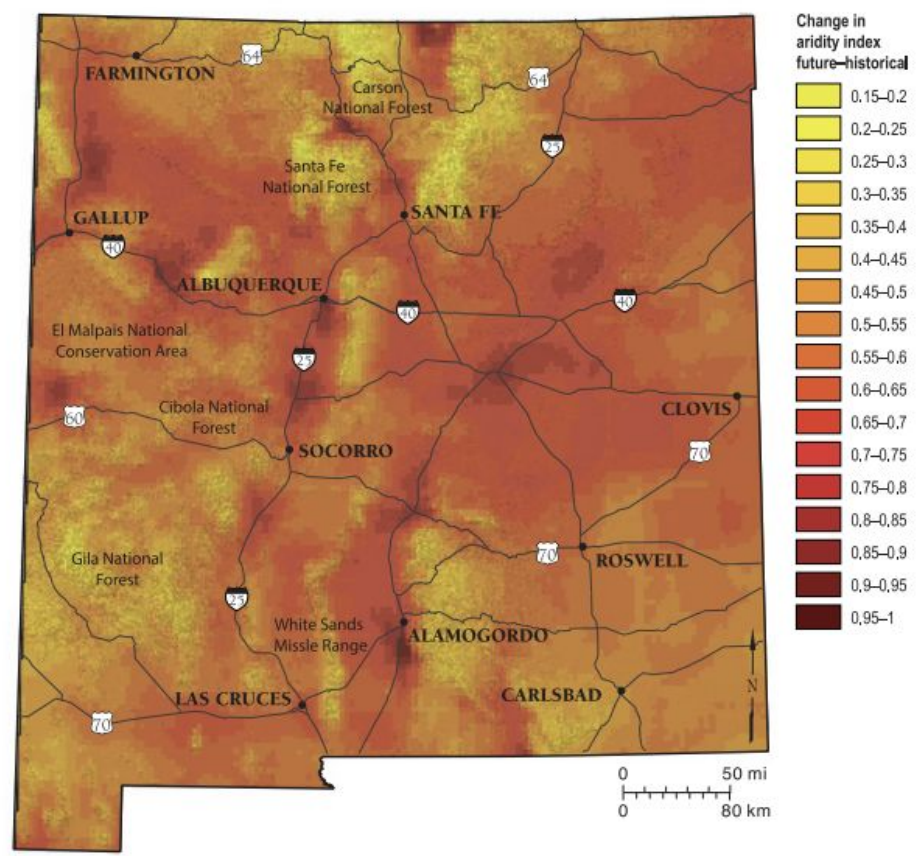
Regional Overview - New Mexico

- Hotter and more intense droughts will lead to decreased surface water
- Therefore many areas of NM will pump groundwater as a replacement which can result in land subsidence.
- Same trend of aridity diminishing surface water will also reduce groundwater recharge
- Shorter and more intense monsoon/rain seasons can lead to flash flooding and more erosion which causes less effective infiltration compared to calm consistent rain
- Most aquifer recharge in NM comes from snowmelt, warmer climate will result in unpredictable and earlier melt seasons which reduces reliable recharge input

"Climate Change and New Mexico's Water Resources: A 50 Year Outlook" NM Bureau of Geology and Mineral Resources



Projected change in water stress for river basins by mid-century (2040–2061) compared to historical average (1900–1970). Brown colors represent higher water stress (generally drier conditions). This figure is borrowed from R. Lindsey in "Climate change to increase water stress in many parts of U.S.," 2013, published by NOAA through www.climate.gov.



Projected difference in local aridity index from 1970–2000 to 2040–2069. Aridity index is the ratio of average potential evapotranspiration to average precipitation. The darker red colors represent the areas in which aridity will increase the most.

References

- https://www.nmlegis.gov/handouts/WNR%20072522%20Item%205%20Gutzler1.pdf
- Impacts of groundwater pumping and climate variability on groundwater availability in the Rio Grande Basin - Sheng - 2013 - Ecosphere - Wiley Online Library
- Global groundwater warming due to climate change | Nature Geoscience